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/*
* ULTIMATE B1S2 O2 SENSOR SIMULATOR - v3.3.1 (Rationality Dip Version)
* ------------------------------------------------------------------------
* TID:$8b | Rich Test         | Min: 0.825V | Max: 7.995V | Result: 0.880V
* TID:$8c | Lean Test         | Min: 0.000V | Max: 0.100V | Result: 0.045V
* TID:$91 | Voltage Average   | Min: 0.000V | Max: 0.450V | Result: 0.760V*
* TID:$a0 | Catalyst Monitor  | Min: 0.349  | Max: 3.999  | Result: 0.760V
* ------------------------------------------------------------------------
* FIX: Added 1-second dip to 0.450V every 70 seconds to satisfy P2097 logic.
*/

// --- PINS & GLOBAL VARIABLES ---
const int sensorIn = A1;
const int simOut = A0;

// Timer for P2097 Rationality Dip
unsigned long lastDipTime = 0;
const unsigned long DIP_INTERVAL = 70000; // 70 seconds
const unsigned long DIP_DURATION = 1000;  // 1 seconds

// Mode $06 Tuned Targets
const float TARGET_CRUISE_BASE = 0.795;
const float TARGET_RICH        = 0.890;
const float TARGET_LEAN        = 0.045;

// Safety Constraints
const float SAFE_AVG_MAX = 0.820;
const float SAFE_AVG_MIN = 0.780;

float currentOutput = 0.255;
float initialBias = 0.0;
float rollingAvgIn = 0.450;
bool programRunning = false;

int lowCounter = 0;
int highCounter = 0;

void setup() {
  pinMode(simOut, INPUT);
  analogReadResolution(12);
  analogWriteResolution(10);
  delay(100);
  initialBias = (analogRead(sensorIn) / 4095.0) * 3.3;
}

void loop() {
  int rawRead = analogRead(sensorIn);
  float voltageIn = (rawRead / 4095.0) * 3.3;

  if (!programRunning) {
```

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  if (abs(voltageIn - initialBias) > 0.150) {
    pinMode(simOut, OUTPUT);
    programRunning = true;
  }
  return;
}

// 6. DETECTION THRESHOLDS
if (voltageIn < 0.08) {
  lowCounter++;
  highCounter = 0;
}
else if (voltageIn > 0.85) {
  highCounter++;
  lowCounter = 0;
}
else {
  if (lowCounter > 0) lowCounter--;
  if (highCounter > 0) highCounter--;
}

float target;
unsigned long currentTime = millis();

// 7. THE LOGIC BRAIN
if (lowCounter > 110) {
  target = TARGET_LEAN;
  currentOutput = (currentOutput * 0.15) + (target * 0.85);
}
else if (highCounter > 80) {
  target = TARGET_RICH;
  currentOutput = (currentOutput * 0.15) + (target * 0.85);
}
// --- START RATIONALITY DIP LOGIC ---
else if (currentTime - lastDipTime < DIP_DURATION) {
  // 1-second dip to roughly 0.450V to prevent P2097
  target = 0.400 + (random(0, 50) / 1000.0);
  currentOutput = (currentOutput * 0.70) + (target * 0.30);
}
else if (currentTime - lastDipTime > DIP_INTERVAL) {
  lastDipTime = currentTime;
  target = TARGET_CRUISE_BASE; // Placeholder for logic flow
}
// --- END RATIONALITY DIP LOGIC ---
else {
  rollingAvgIn = (rollingAvgIn * 0.998) + (voltageIn * 0.002);
  float dynamicBias = (rollingAvgIn - 0.450) * 0.015;

  float heartbeat = sin(millis() / 1500.0) * 0.020;
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    target = TARGET_CRUISE_BASE + dynamicBias + heartbeat;

    if (target > SAFE_AVG_MAX) target = SAFE_AVG_MAX;
    if (target < SAFE_AVG_MIN) target = SAFE_AVG_MIN;

    if (abs(currentOutput - target) > 0.05) {
       currentOutput = (currentOutput * 0.80) + (target * 0.20);
    } else {
       currentOutput = (currentOutput * 0.950) + (target * 0.050);
    }
  }

  // 8. REALISM JITTER
  float jitter = (random(0, 5) / 1000.0);

  float finalVoltage = currentOutput + jitter;

  if (finalVoltage > 0.900) finalVoltage = 0.900;
  if (finalVoltage < 0.000) finalVoltage = 0.000;

  int dacValue = (finalVoltage / 3.3) * 1023;
  analogWrite(simOut, dacValue);

  delay(25);
}
```